PYTHON FOR DATA SCIENCE · LESSON 7 OF 29

# Tuples & Sets: Immutable Records, Unpacking and Set Algebra

Use tuples for fixed records and multiple return values, master unpacking and named tuples, and use sets for uniqueness, fast membership tests and set algebra (union, intersection, difference) on real customer data.

Python class 5 | Python 3.13 | Beginner | 90 min

**By the end of this lesson you can**

- Create tuples, explain immutability and hashability, and unpack tuples including starred unpacking
- Use namedtuple and dataclass-style records for readable fixed structures
- Create sets and frozensets and use add, discard, remove, update and set operators
- Apply union, intersection, difference and symmetric difference to compare groups
- Choose between list, tuple, set and dict for a given task

**Datasets used**

`superstore_sales.csv`

## Tuples

A **tuple** is an ordered, **immutable** sequence written with commas (brackets are usually optional).

Python

```
point = (28.61, 77.20)
single = (5,)                    # the trailing comma makes a one-item tuple
empty = ()
rec = "E101", "Kavya Iyer", "HR", 97000        # packing: no brackets needed
print(point, type(single).__name__, type((5)).__name__, rec[1], rec[-1], rec[1:3],
len(rec))
print(rec.count("HR"), rec.index("HR"), (1, 2) + (3,), (0,) * 3, 5 in (1, 5, 9))
try: rec[0] = "E102"
except TypeError as e: print("Error:", e)
```

OUTPUT

```
(28.61, 77.2) tuple int Kavya Iyer 97000 ('Kavya Iyer', 'HR') 4
1 2 (1, 2, 3) (0, 0, 0) True
Error: 'tuple' object does not support item assignment
```

## Unpacking

Python

```python
emp_id, name, dept, salary = rec                # one name per item
a, b = 1, 2; a, b = b, a                          # swap
first, *others = (10, 20, 30, 40)                 # * collects the rest into a list
*head, last = (10, 20, 30, 40)
x, (y, z) = 1, (2, 3)                             # nested unpacking
_, dept_only, *_ = rec                            # _ means "ignore"
print(emp_id, name, a, b, first, others, head, last, x, y, z, dept_only)
for city, sales in [("Pune", 120), ("Delhi", 95)]:     # unpack in loops
    print(city, sales)
```

OUTPUT

```
E101 Kavya Iyer 2 1 10 [20, 30, 40] [10, 20, 30] 40 1 2 3 Kavya Iyer
Pune 120
Delhi 95
```

## Why tuples?

| Use | Example |
|---|---|
| **Fixed records** | `(lat, lon)`, `(year, month)`, a row from a database |
| **Multiple return values** | `def stats(x): return min(x), max(x)` → `lo, hi = stats(data)` |
| **Dictionary keys and set members** (hashable) | `sales[("North", "Technology")]` |
| **Protection** from accidental change | Constants like `METROS = ("Delhi", "Mumbai")` |
| **Slightly less memory and faster** than lists | |

Python

```
def describe(values):
    return min(values), max(values), sum(values) / len(values)
lo, hi, avg = describe([4, 8, 15, 16, 23, 42])
print(lo, hi, round(avg, 2), describe([1, 2]))
print(hash((1, 2)), "| a tuple containing a list is not hashable:", end=" ")
try: hash((1, [2]))
except TypeError as e: print(e)
```

OUTPUT

```
4 42 18.0 (1, 2, 1.5)
-3550055125485641917 | a tuple containing a list is not hashable: unhashable type:
'list'
```

**NOTE** **Tuples are immutable, but their contents might not be**

`t = ([1, 2], 3)`; `t[0].append(9)` works, because the tuple holds a **reference** to a mutable list. The tuple itself still points to the same list.

## Named tuples and data classes

When a record has many fields, positions like `rec[3]` are hard to read. Name the fields:

Python

```
from collections import namedtuple
from dataclasses import dataclass
Employee = namedtuple("Employee", ["emp_id", "name", "dept", "salary"])
e = Employee("E101", "Kavya Iyer", "HR", 97000)
print(e, e.name, e[3], e._asdict(), e._replace(salary=100000).salary)

@dataclass(frozen=True)                # frozen = immutable, like a tuple, but with types
and methods
class Product:
    sku: str
    price: float
    def with_gst(self, rate=0.18): return round(self.price * (1 + rate), 2)
p = Product("P101", 4500.0)
print(p, p.with_gst())
```

OUTPUT

```
Employee(emp_id='E101', name='Kavya Iyer', dept='HR', salary=97000) Kavya Iyer 97000
{'emp_id': 'E101', 'name': 'Kavya Iyer', 'dept': 'HR', 'salary': 97000} 100000
Product(sku='P101', price=4500.0) 5310.0
```

## Sets

A **set** is an **unordered** collection of **unique**, hashable items, written with braces. Sets support fast membership tests and mathematical set operations.

Python

```python
cities = {"Pune", "Delhi", "Kochi", "Delhi"}          # duplicates vanish
print(cities, len(cities), "Pune" in cities)
empty = set()                                          # {} is an empty DICT, not a set!
print(set("mississippi"), set([3, 1, 2, 3, 1]), sorted({5, 2, 9}), type({}).__name__,
type(empty).__name__)
```

OUTPUT

```
{'Delhi', 'Pune', 'Kochi'} 3 True
{'m', 'p', 's', 'i'} {1, 2, 3} [2, 5, 9] dict set
```

| Method / operator | Effect |
|---|---|
| `s.add(x)` | Add an item |
| `s.update(iterable)` or `s \|= other` | Add many items |
| `s.discard(x)` | Remove x if present (no error) |
| `s.remove(x)` | Remove x; `KeyError` if missing |
| `s.pop()` | Remove and return an arbitrary item |
| `s.clear()` | Empty the set |
| `x in s` | Membership, **O(1)** on average |
| `len(s)` , `min(s)` , `max(s)` , `sum(s)` , `sorted(s)` | Usual helpers (sorted returns a list) |

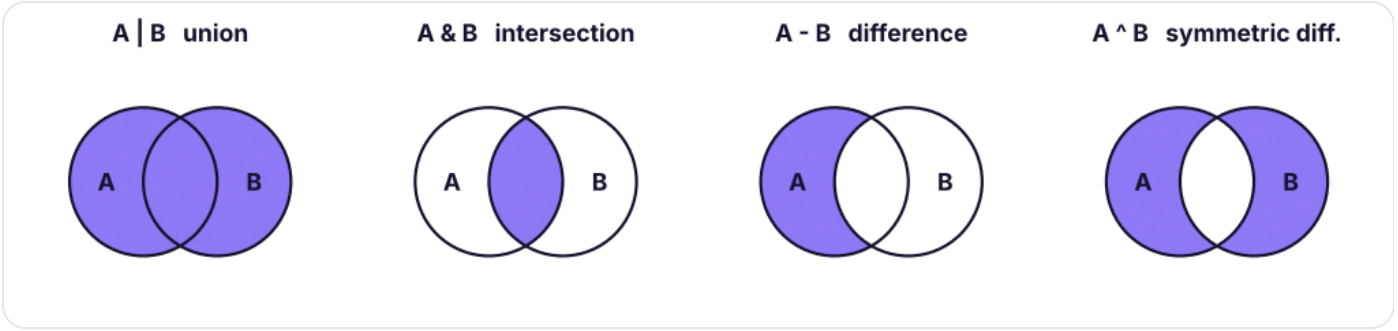
Python

```python
s = {1, 2, 3}
s.add(4); s.update([5, 6]); s.discard(10); s.remove(1)
print(s, s.pop() in {2, 3, 4, 5, 6})
```

OUTPUT

```
{3, 4, 5, 6} True
```

## Set algebra




The four set operations as Venn diagrams (A and B are sets of customers).

Python

```python
a = {"Asha", "Ravi", "Meera", "Kabir"}
b = {"Meera", "Kabir", "Zoya", "Neha"}
print("union       :", sorted(a | b), "  also a.union(b)")
print("intersection:", sorted(a & b), "  also a.intersection(b)")
print("A - B       :", sorted(a - b), "  also a.difference(b)")
print("B - A       :", sorted(b - a))
print("A ^ B       :", sorted(a ^ b), "  (in exactly one)")
print(a <= {"Asha", "Ravi", "Meera", "Kabir", "X"}, {"Asha"} < a, a.isdisjoint({"Zoya"}))
# subset, proper subset, disjoint
```

OUTPUT

```
union       : ['Asha', 'Kabir', 'Meera', 'Neha', 'Ravi', 'Zoya']   also a.union(b)
intersection: ['Kabir', 'Meera']   also a.intersection(b)
A - B       : ['Asha', 'Ravi']   also a.difference(b)
B - A       : ['Neha', 'Zoya']
A ^ B       : ['Asha', 'Neha', 'Ravi', 'Zoya']   (in exactly one)
True True True
```

These are the same operations as SQL's `UNION`, `INTERSECT` and `EXCEPT` (SQL Lesson 9).

## Frozen sets

`frozenset` is an immutable set: it can be a dictionary key or a member of another set.

Python

```python
fs = frozenset({"North", "South"})
print(fs, {fs: "regions"}, fs | {"East"})
```

OUTPUT

```
frozenset({'North', 'South'}) {frozenset({'North', 'South'}): 'regions'}
frozenset({'North', 'South', 'East'})
```

## Real data: customer movement with sets

The Superstore data lets us rebuild the retention analysis from SQL Lesson 9 with sets:

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
cust = {y: set(s.loc[s.order_date.dt.year == y, "customer_id"]) for y in (2023, 2024,
2025)}
print({y: len(c) for y, c in cust.items()})
retained = cust[2024] & cust[2025]
lost = cust[2024] - cust[2025]
new_in_2025 = cust[2025] - (cust[2023] | cust[2024])
print(f"Retained 2024→2025: {len(retained)} | lost: {len(lost)} | brand new in 2025:
{len(new_in_2025)}")
print(f"Retention rate: {len(retained) / len(cust[2024]):.1%}  (matches the SQL result: 467
of 590)")
print("Bought in all 3 years:", len(cust[2023] & cust[2024] & cust[2025]), "| bought in
exactly one year:", len((cust[2023] ^ cust[2024] ^ cust[2025]) - (cust[2023] & cust[2024] &
cust[2025])))
```

OUTPUT

```
{2023: 540, 2024: 590, 2025: 623}
Retained 2024→2025: 467 | lost: 123 | brand new in 2025: 52
Retention rate: 79.2%  (matches the SQL result: 467 of 590)
Bought in all 3 years: 311 | bought in exactly one year: 128
```

Same answers as the SQL queries, in three lines. Sets are also the fastest way to **de-duplicate** and to **check membership** in big collections:

Python

```
import time, random
big_list = list(range(200_000)); big_set = set(big_list); probes =
[random.randrange(400_000) for _ in range(300)]
t0 = time.perf_counter(); [p in big_list for p in probes]; t1 = time.perf_counter(); [p in
big_set for p in probes]; t2 = time.perf_counter()
print(f"300 lookups: list {1000 * (t1 - t0):.1f} ms | set {1000 * (t2 - t1):.3f} ms")
```

OUTPUT

```
300 lookups: list 329.5 ms | set 0.244 ms
```

## Choosing a collection

| Need | Use |
|---|---|
| Ordered, changeable sequence | `list` |
| Ordered, fixed record or dict key | `tuple` |
| Unique items, fast "is it in there?", set algebra | `set` |
| Key → value lookup | `dict` (next lesson) |
| Unique **and** ordered | `dict.fromkeys(items)` or a list + set |
| FIFO queue | `collections.deque` |
| Counting | `collections.Counter` |

## Summary

**KEY TAKEAWAYS**

- **Tuples** are ordered and immutable: use them for fixed records, multiple return values and dictionary keys; unpack with `a, b = t` and `first, *rest = t`.
- A one-item tuple needs a comma: `(5,)`. Use **namedtuple** or a frozen **dataclass** for readable records.
- **Sets** hold unique, hashable items with O(1) membership; `{}` is an empty dict, so use `set()`.
- Set algebra: `|` union, `&` intersection, `-` difference, `^` symmetric difference; subset tests with `<=`.
- Sets turn retention/overlap analyses into one-liners and are far faster than lists for membership tests.

### Quick quiz

6 questions · tap an answer to check it

**1.** What is `type((5))` ?

**A** tuple

**B** int

**C** list

**D** set

**Answer: B.** Brackets alone do not make a tuple; `(5,)` does.

**2.** What does `{1, 2, 3} & {2, 3, 4}` return?

**A** {1, 2, 3, 4}

**B** {2, 3}

**C** {1, 4}

**D** {1}

**Answer: B.** `&` is the intersection.

**3.** Which can be used as a dictionary key?

**A** A list

**B** A tuple of strings

**C** A set

**D** A dict

**Answer: B.** Keys must be hashable; lists, sets and dicts are not.

**4.** How do you create an empty set?

A `{}`

B `set()`

C `[]`

D `()`

**Answer: B.** `{}` creates an empty dictionary.

**5.** What does `first, *rest = (1, 2, 3, 4)` assign to `rest`?

A (2, 3, 4)

B [2, 3, 4]

C 4

D 2

**Answer: B.** Starred unpacking collects the remaining items into a list.

**6.** Why are set lookups faster than list lookups?

A Sets are sorted

B Sets use hashing, giving O(1) average membership tests

C Sets store fewer items

D They are not faster

**Answer: B.** A list scans item by item (O(n)).

EXERCISE

### Set problems

(1) Find the common cities between `{"Pune","Delhi","Goa"}` and `{"Goa","Kochi","Pune"}`. (2) Customers who bought in 2023 and 2024 but not in 2025 (use the Superstore sets). (3) Write `min_max_mean(values)` returning a tuple, and unpack it. (4) Check whether two words are anagrams using `sorted` and also using `collections.Counter`.

**Show solution (worked answer)**

Python

```
print({"Pune","Delhi","Goa"} & {"Goa","Kochi","Pune"})
print(len((cust[2023] & cust[2024]) - cust[2025]))
def min_max_mean(v): return min(v), max(v), sum(v) / len(v)
lo, hi, m = min_max_mean([3, 9, 6])
from collections import Counter
print(sorted("listen") == sorted("silent"), Counter("listen") == Counter("silent"))
```

INTERVIEW CORNER

### Questions you may be asked

**Q: Tuple vs list: when would you choose each?** Tuple for fixed, heterogeneous records, hashable keys and returning several values; list for homogeneous collections that change.

**Q: How do you remove duplicates from a list while preserving order?** `list(dict.fromkeys(items))`; `list(set(items))` does not preserve order.

**Q: What is the time complexity of set membership and why?** Average O(1) because sets are hash tables (worst case O(n) with many collisions).

**Q: How do you find items in A but not in B?** `A - B` for sets; for lists use a set for B: `[x for x in A if x not in set_b]`.

## Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.